

進階機器學習

Homework #5

Due 2025 May 14 11:00 pm

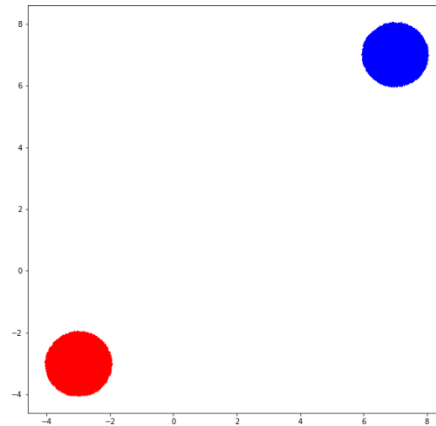
(一) Use the standard MNIST training dataset as the normal data. The test dataset should include both MNIST test data and Fashion-MNIST test data. Implement the following steps for abnormal detection:

1. Train an image classifier using the normal data (MNIST training set). Use this classifier to detect anomalies by identifying Fashion-MNIST images in the test dataset.
2. Train a standard autoencoder on the normal data. Use its reconstruction error as a basis for detecting anomalies in the test dataset.
3. Train a denoising autoencoder using the normal data, and again use it to detect anomalies based on reconstruction error.
4. Apply the Isolation Forest algorithm directly on the data to identify Fashion-MNIST images as anomalies in the test dataset.
5. Repeat step 4, but first extract features from the data using a pre-trained model (e.g., for MNIST digits) before applying the Isolation Forest.
6. Compare and analyze the anomaly detection performance of all five methods. Determine and justify an appropriate threshold for each method.

(二) Consider the use of autoencoder:

1. Select one image each of digits 0, 1, 6, and 7 from the MNIST dataset. Use these four images to train both a standard autoencoder and a variational autoencoder (VAE), with the latent space dimension set to 1.
 - i. Show the latent codes (encodings) of these four images for both models.
 - ii. Generate and compare the decoder outputs by feeding various code values into each decoder, and visualize the differences in generation behavior between the two models.
2. Train a VAE using the entire MNIST training dataset, setting the latent space dimension to 2.
 - i. Visualize the distribution of the encoded mean and variance for all images. Use different colors to represent different digit classes. Show three separate cases: (a) Using only reconstruction loss (b) Using only KL divergence loss (c) Using both losses together.
 - ii. The KL divergence loss measures how much the latent distribution deviates from a standard normal distribution. Discuss the potential impact on the model behavior and latent space if the prior distribution is changed from a normal distribution to a uniform distribution.

(三) Suppose we have two clusters of 2D data, each distributed within a circular region centered at $(7, 7)$ and $(-3, -3)$, respectively, as illustrated in the figure below.



1. Use one of the following generative models—GAN, WGAN, or WGAN-GP—to learn and replicate the data distribution.
 - i. Generate 50 data points using the trained model.
 - ii. Plot the generated points on a 2D graph to visualize how well they match the original distribution.

2. Use InfoGAN with an additional categorical input to learn disentangled representations of the two data clusters.
 - i. Control the categorical input to specify which group (cluster) of data to generate.
 - ii. Plot and demonstrate how the model can generate data from each specific cluster by manipulating the categorical variable.